

Solutions – Quiz 6

ECON 320 – Introduction to Mathematical Economics

Instructor: Muhebullah Karimzada

December 01, 2025

Part A — Multiple Choice (1 mark each)

For each question, we give the correct option and a brief explanation.

1. **Correct answer: (c)**

The Envelope Theorem says that when computing the effect of a parameter on the *optimized* value of the objective, the indirect effect via optimal choice variables drops out because the first-order conditions (FOCs) hold. What remains is the *direct* partial derivative. Hence, only the direct partial effect matters; indirect effects cancel at the optimum.

2. **Correct answer: (b)**

In constrained optimization, the Lagrange multiplier λ measures the change in the maximized value of the objective when the constraint right-hand side is relaxed by one unit. This is the *shadow value* of the constraint. So (b) is correct.

3. **Correct answer: (a)**

The *indirect utility function* $V(p, M)$ gives the maximum utility as a function of prices and income. The *expenditure function* $E(p, U)$ gives the minimum income required to achieve a given utility level. These are dual to each other: one maximizes utility given a budget, the other minimizes expenditure given a utility target.

4. **Correct answer: (b)**

Shephard's Lemma states that if $C(w, r, Q)$ is the cost function (obtained from cost minimization), then:

$$\frac{\partial C}{\partial w} = L^*(w, r, Q), \quad \frac{\partial C}{\partial r} = K^*(w, r, Q),$$

i.e., the partial derivatives of cost with respect to input prices equal the conditional factor demands.

5. **Correct answer: (c)**

$\int_0^T I(t) dt$ is the continuous-time sum (integral) of the investment flow $I(t)$ between 0 and T . It therefore represents the *total accumulated investment* over $[0, T]$.

6. **Correct answer: (b)**

A first-order linear differential equation has the standard form:

$$\frac{dx}{dt} + P(t)x = Q(t),$$

where $P(t)$ and $Q(t)$ are given functions of t . Option (b) matches this.

7. **Correct answer: (d)**

At a steady state of

$$\frac{dK}{dt} = I - \delta K,$$

we set $\frac{dK}{dt} = 0$, giving:

$$0 = I - \delta K^* \quad \Rightarrow \quad K^* = \frac{I}{\delta}.$$

Part B — Short Answer Problems

Question 1 — Envelope Theorem (4 marks)

We have:

$$\max_{x,y} U(x, y, a) \quad \text{s.t.} \quad g(x, y, a) = 0,$$

with Lagrangian:

$$\mathcal{L}(x, y, \lambda, a) = U(x, y, a) + \lambda g(x, y, a).$$

Let $x^*(a), y^*(a), \lambda^*(a)$ denote the optimal choices as functions of a , and let the value function be

$$V(a) = U(x^*(a), y^*(a), a).$$

(1) Derive $\frac{dV}{da}$ using the Envelope Theorem. **Step 1:** Differentiate $V(a)$ by the chain rule:

$$\frac{dV}{da} = \frac{\partial U}{\partial x} \frac{dx^*}{da} + \frac{\partial U}{\partial y} \frac{dy^*}{da} + \frac{\partial U}{\partial a}.$$

All partial derivatives of U are evaluated at $(x^*(a), y^*(a), a)$.

Step 2: Use the first-order conditions (FOCs) from the Lagrangian:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= U_x + \lambda g_x = 0, \\ \frac{\partial \mathcal{L}}{\partial y} &= U_y + \lambda g_y = 0, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= g(x, y, a) = 0. \end{aligned}$$

From the first two:

$$U_x = -\lambda g_x, \quad U_y = -\lambda g_y.$$

Step 3: Substitute into the expression for $\frac{dV}{da}$:

$$\frac{dV}{da} = (-\lambda g_x) \frac{dx^*}{da} + (-\lambda g_y) \frac{dy^*}{da} + U_a.$$

Factor out $-\lambda$ in the first two terms:

$$\frac{dV}{da} = U_a - \lambda \left(g_x \frac{dx^*}{da} + g_y \frac{dy^*}{da} \right).$$

Step 4: Total differentiate the constraint:

$$g(x^*(a), y^*(a), a) = 0.$$

Then:

$$\frac{dg}{da} = g_x \frac{dx^*}{da} + g_y \frac{dy^*}{da} + g_a = 0.$$

So:

$$g_x \frac{dx^*}{da} + g_y \frac{dy^*}{da} = -g_a.$$

Step 5: Substitute this into the expression for $\frac{dV}{da}$:

$$\frac{dV}{da} = U_a - \lambda(-g_a) = U_a + \lambda g_a.$$

Thus the **Envelope Theorem** gives:

$$\boxed{\frac{dV}{da} = \frac{\partial U}{\partial a} + \lambda \frac{\partial g}{\partial a}}$$

evaluated at the optimum.

(If the constraint is written as $k - g(x, y, a) = 0$, the formula becomes $dV/da = U_a - \lambda g_a$.)

(2) Why do the indirect effects through x^*, y^* drop out? At the optimum, the FOCs imply $U_x = -\lambda g_x$ and $U_y = -\lambda g_y$. When we differentiate $V(a)$, the terms involving dx^*/da and dy^*/da appear multiplied by U_x and U_y , but these are replaced by $-\lambda g_x$ and $-\lambda g_y$ and then combined with the total derivative of the constraint. Because the constraint must continue to hold as a changes, its total derivative is zero, which forces the combination

$$g_x \frac{dx^*}{da} + g_y \frac{dy^*}{da}$$

to be exactly $-g_a$. This cancels neatly, leaving only the direct U_a and g_a terms. In other words, the optimal adjustment of x^* and y^* ensures that their indirect contributions vanish in the derivative of the value function.

(3) **Economic interpretation of $-\lambda \frac{\partial g}{\partial a}$.** If we write the constraint as $k - g(x, y, a) = 0$, then:

$$\frac{dV}{da} = U_a - \lambda g_a.$$

The term $-\lambda g_a$ measures how changes in the parameter a tighten or relax the constraint, multiplied by the shadow value λ . Economically, g_a tells us how the constraint changes with a , and λ gives the value of relaxing the constraint. Thus $-\lambda g_a$ is the contribution to the change in the optimized value that comes from the way a changes the feasible set, weighted by the marginal value (shadow price) of the constraint.

Question 2 — Duality and Shephard's Lemma (4 marks)

We have a cost function:

$$C(w, r, Q) = wL^*(Q) + rK^*(Q),$$

where w is wage, r rental rate of capital, and $L^*(Q), K^*(Q)$ are cost-minimizing demands for labor and capital given Q .

(1) **Use Shephard's Lemma to derive factor demands.** To be precise, $C(w, r, Q)$ arises from the *cost minimization problem*:

$$C(w, r, Q) = \min_{L, K} \{wL + rK : F(L, K) \geq Q\}.$$

We form the Lagrangian:

$$\mathcal{L}(L, K, \lambda) = wL + rK + \lambda(Q - F(L, K)),$$

where $\lambda \geq 0$ is the multiplier.

FOCs:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial L} &= w - \lambda F_L(L, K) = 0, \\ \frac{\partial \mathcal{L}}{\partial K} &= r - \lambda F_K(L, K) = 0, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= Q - F(L, K) = 0. \end{aligned}$$

These conditions define $L^*(w, r, Q)$ and $K^*(w, r, Q)$.

By the **Envelope Theorem** applied to the minimization problem with respect to w and r :

$$\frac{\partial C}{\partial w} = \frac{\partial}{\partial w} \mathcal{L}(L^*, K^*, \lambda^*) = L^*,$$

because the derivative of $wL + rK$ with respect to w is L , and all terms involving $\partial L^*/\partial w$, $\partial K^*/\partial w$ drop out via the FOCs.

Similarly:

$$\frac{\partial C}{\partial r} = K^*.$$

Hence:

$$\boxed{\frac{\partial C(w, r, Q)}{\partial w} = L^*(w, r, Q), \quad \frac{\partial C(w, r, Q)}{\partial r} = K^*(w, r, Q).}$$

(2) Why does cost-minimizing behavior imply duality between cost and production? The **primal** production problem is to choose inputs (L, K) to maximize output $F(L, K)$ subject to input constraints. The **dual** cost-minimization problem is, given a target output Q , to choose (L, K) to minimize total cost $wL + rK$ subject to $F(L, K) \geq Q$. Both problems share the same technology $F(L, K)$ and hence share the same tangency condition $MRTS = w/r$. Therefore, for any given Q , the cost-minimizing (L^*, K^*) correspond to the efficient input combination that would arise from the production maximization problem. This one-to-one relationship between the primal and dual problems is the duality in production theory.

(3) Show that marginal cost equals $\partial C/\partial Q$. By definition,

$$MC(Q) = \frac{dC}{dQ},$$

the additional cost of producing one more unit of output.

From the cost-minimization problem, the Lagrange multiplier λ associated with the constraint $F(L, K) \geq Q$ measures the change in minimum cost when we relax the required output level Q by one unit. By the Envelope Theorem with respect to Q :

$$\frac{\partial C}{\partial Q} = \lambda.$$

Economically, λ is the *marginal cost* of output: increasing required output slightly raises

minimal cost by approximately λ per unit. Thus:

$$\boxed{MC(Q) = \frac{\partial C}{\partial Q}.$$

Question 3 — Integration and Stock–Flow Dynamics (4 marks)

We are given:

$$\frac{dK}{dt} = I(t) - \delta K(t),$$

where $I(t)$ is investment and $\delta > 0$ is the depreciation rate.

(1) Integrating factor and solution for $K(t)$. Rewrite the equation in standard linear form:

$$\frac{dK}{dt} + \delta K = I(t).$$

This matches:

$$\frac{dK}{dt} + P(t)K = Q(t),$$

with $P(t) = \delta$ and $Q(t) = I(t)$.

Step 1: Compute the integrating factor:

$$\mu(t) = e^{\int P(t)dt} = e^{\int \delta dt} = e^{\delta t}.$$

Step 2: Multiply both sides of the differential equation by $\mu(t)$:

$$e^{\delta t} \frac{dK}{dt} + \delta e^{\delta t} K = e^{\delta t} I(t).$$

Step 3: Recognize the left-hand side as the derivative of a product:

$$\frac{d}{dt}(e^{\delta t} K(t)) = e^{\delta t} \frac{dK}{dt} + \delta e^{\delta t} K(t).$$

Hence:

$$\frac{d}{dt}(e^{\delta t} K(t)) = e^{\delta t} I(t).$$

Step 4: Integrate both sides from 0 to t :

$$\int_0^t \frac{d}{ds}(e^{\delta s} K(s))ds = \int_0^t e^{\delta s} I(s)ds.$$

Left-hand side:

$$e^{\delta t} K(t) - e^{\delta \cdot 0} K(0) = e^{\delta t} K(t) - K(0).$$

Thus:

$$e^{\delta t} K(t) - K(0) = \int_0^t e^{\delta s} I(s) ds.$$

Step 5: Solve for $K(t)$ by dividing through by $e^{\delta t}$:

$$K(t) = e^{-\delta t} K(0) + e^{-\delta t} \int_0^t e^{\delta s} I(s) ds.$$

Equivalently (sometimes written in convolution form):

$$K(t) = K(0)e^{-\delta t} + \int_0^t e^{-\delta(t-s)} I(s) ds.$$

(2) Long-run (steady-state) value of K . A steady state K^* is a constant level such that $\frac{dK}{dt} = 0$ and the system is in balance. Suppose $I(t) = \bar{I}$ is constant in the long run. Then the dynamic equation becomes:

$$\frac{dK}{dt} = \bar{I} - \delta K.$$

At steady state, $\frac{dK}{dt} = 0$, so:

$$0 = \bar{I} - \delta K^* \quad \Rightarrow \quad K^* = \frac{\bar{I}}{\delta}.$$

(3) Economic interpretation of $K(0)e^{-\delta t}$ and $\frac{1}{\delta} \int_0^t e^{-\delta(t-s)} I(s) ds$. From the solution

$$K(t) = K(0)e^{-\delta t} + \int_0^t e^{-\delta(t-s)} I(s) ds,$$

we can interpret the terms as follows:

- $K(0)e^{-\delta t}$: This is the original capital stock decaying exponentially at rate δ . Over time, if no new investment occurred, the capital stock would shrink toward zero due to depreciation.
- $\int_0^t e^{-\delta(t-s)} I(s) ds$: This term accumulates all past investments $I(s)$, each discounted by $e^{-\delta(t-s)}$ to reflect depreciation between the time of investment s and the current time t .

t . Recent investments (with $t - s$ small) are discounted less, while older investments are discounted more because they have had more time to depreciate. This integral therefore represents the *effective capital* contributed by all past investment flows, after accounting for depreciation.

Question 4 — Present Value Calculation (4 marks)

Income process:

$$Y(t) = Y_0 e^{gt},$$

with constant growth rate g and initial level $Y_0 > 0$. Discount rate is $r > g$.

We want:

$$PV = \int_0^\infty Y(t) e^{-rt} dt = \int_0^\infty Y_0 e^{gt} e^{-rt} dt.$$

(1) Compute the present value (step-by-step). Step 1: Combine the exponentials:

$$Y_0 e^{gt} e^{-rt} = Y_0 e^{(g-r)t}.$$

So:

$$PV = \int_0^\infty Y_0 e^{(g-r)t} dt = Y_0 \int_0^\infty e^{(g-r)t} dt.$$

Step 2: Let $k = g - r$. Then:

$$PV = Y_0 \int_0^\infty e^{kt} dt.$$

To integrate, first consider the finite upper limit T :

$$\int_0^T e^{kt} dt = \left[\frac{1}{k} e^{kt} \right]_0^T = \frac{1}{k} e^{kT} - \frac{1}{k}.$$

Step 3: Take the limit as $T \rightarrow \infty$.

We have two cases depending on the sign of k :

- If $k < 0$ (i.e., $g - r < 0$ or $r > g$), then $e^{kT} \rightarrow 0$ as $T \rightarrow \infty$, and:

$$\lim_{T \rightarrow \infty} \left(\frac{1}{k} e^{kT} - \frac{1}{k} \right) = 0 - \frac{1}{k} = -\frac{1}{k}.$$

Since $k = g - r$, we have $-1/k = -1/(g - r) = 1/(r - g)$.

- If $k \geq 0$ (i.e., $r \leq g$), then e^{kT} does not go to zero and the integral diverges.

Therefore, when $r > g$:

$$\int_0^\infty e^{(g-r)t} dt = \frac{1}{r-g}.$$

Step 4: Multiply by Y_0 :

$$PV = Y_0 \cdot \frac{1}{r-g} = \boxed{\frac{Y_0}{r-g}}.$$

(2) Condition for finiteness and explanation. The integral:

$$\int_0^\infty e^{(g-r)t} dt$$

converges only if the exponent $(g-r)t$ goes to $-\infty$ as $t \rightarrow \infty$, which requires $g-r < 0$, i.e.,

$$\boxed{r > g}.$$

If $r > g$, the discount factor e^{-rt} dominates the growth e^{gt} so that the product $e^{(g-r)t}$ tends to zero and the present value is finite. If $r \leq g$, the effective growth of $Y(t)e^{-rt}$ is non-negative, causing the integral (present value) to diverge: future income grows too fast (or is insufficiently discounted), and its present value becomes infinite.